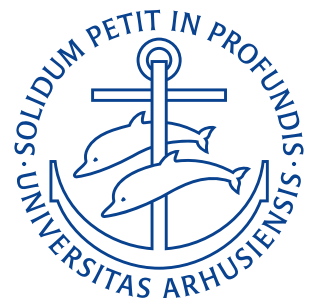


Department of Mechanical and Production Engineering
Aarhus University

Assignment 2

Author: Noah Rahbek Bigum Hansen
Student ID: 202405538
Course: Electromagnetism & Electronics — 290231U007
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Problem 1A

A parallel-plate air-filled capacitor having area 40 cm^2 and plate spacing $1,0 \text{ mm}$ is charged to a potential difference of 500 V . Find:

- the capacitance,
- the magnitude of the charge on each plate,
- the stored energy,
- the electric field between the plates, and
- the energy density between the plates.

Assumptions

- The air can be modelled as a vacuum ($\kappa = 1,00054 \approx 1$)

Givens

$A = 40 \text{ cm}^2$	Area of capacitor plates
$d = 1,0 \text{ mm}$	Plate spacing
$V = 500 \text{ V}$	Potential difference

Derivation

- The capacitance of a parallel-plate capacitor is

$$C = \epsilon_0 \frac{A}{d}.$$

Plugging in known values we get:

$$C = 8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2 \cdot \frac{40 \text{ cm}^2}{1,0 \text{ mm}} = 35,4 \text{ pF}.$$

- The capacitance is defined as

$$C = \frac{Q}{V} \implies Q = CV.$$

Where Q is the charge and V the voltage difference. Hence,

$$Q = 35,4 \text{ pF} \cdot 500 \text{ V} = 17,7 \text{ nC}.$$

- The stored electrical energy in a capacitor is

$$U_E = \frac{1}{2} CV^2.$$

Hence,

$$U_E = \frac{1}{2} \cdot 35,4 \text{ pF} \cdot (500 \text{ V})^2 = 4,425 \mu\text{J}.$$

- The potential difference of a plate capacitor is the product of the internal electric field and the separation distance:

$$V = Ed \implies E = \frac{V}{d} = \frac{500 \text{ V}}{1,0 \text{ mm}} = 500 \cdot 10^3 \frac{\text{N}}{\text{C}}.$$

- The energy stored in the capacitor was found to be $U_E = 4,425 \mu\text{J}$ above. The volume inside the capacitor is

$Ad = 40\text{ cm}^2 \cdot 1,00\text{ mm}$, hence, the energy per unit volume is:

$$u_E = \frac{U_E}{Ad} = \frac{4,425\mu\text{J}}{40\text{ cm}^2 \cdot 1,00\text{ mm}} = 1,11\text{ J/m}^3.$$

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|-----|--|
| (a) | $C = 35,4\text{ pF}$ |
| (b) | $Q = 17,7\text{ nC}$ |
| (c) | $U_E = 4,425\mu\text{J}$ |
| (d) | $E = 500 \cdot 10^3 \frac{\text{N}}{\text{C}}$ |
| (e) | $u_E = 1,11\text{ J/m}^3.$ |

RESULT

Problem 2A

A dielectric material is to fill the space in a capacitor. Initially, with only air in place, the capacitance is 8,0 pF. With the dielectric material in place, the capacitor should store 3,2 μJ at a maximum potential difference of 350,8 V.

- What dielectric constant is required?
- Of the materials in the Table on the right, which material should be used?

Material	Dielectric Constant κ	Dielectric Strength (kV/mm)
Air (1 atm)	1.00054	3
Polystyrene	2.6	24
Paper	3.5	16
Transformer oil	4.5	
Pyrex	4.7	14
Ruby mica	5.4	
Porcelain	6.5	
Silicon	12	
Germanium	16	
Ethanol	25	
Water (20°C)	80.4	
Water (25°C)	78.5	
Titania ceramic	130	
Strontium titanate	310	8
For a vacuum, $\kappa = \text{unity}$.		

Assumptions

- The air can be modelled as a vacuum ($\kappa = 1,00054 \approx 1$)

Givens

$C_0 = 8,0 \text{ pF}$	Capacitance of the capacitor with air inside
$U_E = 3,2 \mu\text{J}$	Required energy storage of capacitor
$V = 350,8 \text{ V}$	Potential difference across capacitor

Derivation

- The potential energy U_E stored in a capacitor of capacitance C with voltage difference V is

$$U_E = \frac{1}{2} C V^2.$$

Solving for the required capacitance we get:

$$C = \frac{2U_E}{V^2} = \frac{2 \cdot 3,2 \mu\text{J}}{(350,8 \text{ V})^2} = 52 \text{ pF}.$$

Inserting a dielectric in the capacitor raises the vacuum capacitance of the capacitor by a factor κ :

$$C = \kappa C_0 \implies \kappa = \frac{C}{C_0} = \frac{52 \text{ pF}}{8,0 \text{ pF}} = 6,5.$$

- Porcelain has a dielectric constant of $\kappa = 6,5$, meaning it is the closest match.

- (a) $\kappa = 6,5$
 (b) Porcelain.

RESULT

Problem 3A

A coil is connected in series with a $23\text{ k}\Omega$ resistor. An ideal $50,0\text{ V}$ battery is applied across the two devices, and the current reaches a value of $2,00\text{ mA}$ after $5,00\text{ ms}$.

- Find the inductance of the coil.
- How much energy is stored in the coil at this same moment?

Givens

$R = 23\text{ k}\Omega$	Resistance of resistor
$V_s = 50,0\text{ V}$	Potential difference of battery
$I(5,00\text{ ms}) = 2,00\text{ mA}$	Current at $t = 5,00\text{ ms}$

Derivation

- The current in an inductor follows

$$I(t) = \frac{V_s}{R} \left(1 - e^{-\frac{tR}{L}} \right).$$

Solving this for the inductance L we get:

$$\begin{aligned} \frac{RI(t)}{V_s} &= 1 - e^{-\frac{tR}{L}} \\ e^{-\frac{tR}{L}} &= 1 - \frac{RI(t)}{V_s} \\ \frac{tR}{L} &= -\ln\left(1 - \frac{RI(t)}{V_s}\right) \\ \frac{tR}{L} &= \ln\left(\frac{1}{1 - \frac{RI(t)}{V_s}}\right) \\ L &= \frac{tR}{\ln\left(\frac{1}{1 - \frac{RI(t)}{V_s}}\right)} \\ &= \frac{5,00\text{ ms} \cdot 23\text{ k}\Omega}{\ln\left(\frac{1}{1 - \frac{23\text{ k}\Omega \cdot 2,00\text{ mA}}{50,0\text{ V}}}\right)} \\ &= 45\text{ H}. \end{aligned}$$

- The energy stored in an inductor is:

$$U_L = \frac{1}{2} LI^2.$$

Using $L = 45\text{ H}$ and $I(5,00\text{ ms}) = 2,00\text{ mA}$ we get:

$$U_L = \frac{1}{2} \cdot 45\text{ H} \cdot (2,00\text{ mA})^2 = 0,09\text{ mJ}.$$

- | | |
|-----|-------------------------|
| (a) | $L = 45\text{ H}$ |
| (b) | $U_L = 0,09\text{ mJ}.$ |

RESULT

Problem 1B

Find the average and rms value of $x(t) = 2 \cos(\omega t + \pi/3) + 6$.

Derivation

For a periodic function $x(t)$ with period T , the average value $\langle x(t) \rangle$ is

$$\langle x(t) \rangle = \frac{1}{T} \int_0^T x(t') dt'.$$

Inserting known values we get:

$$\begin{aligned} \langle x(t) \rangle &= \frac{1}{T} \int_0^T \left(2 \cos\left(\omega t + \frac{\pi}{3}\right) + 6 \right) dt \\ &= \frac{1}{T} \left(2 \int_0^T \cos\left(\omega t + \frac{\pi}{3}\right) dt + 6 \int_0^T dt \right) \\ &= \frac{1}{T} (2 \cdot 0 + 6T) \\ &= 6. \end{aligned}$$

The rms value is

$$x_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t') dt'}.$$

We let $\alpha = \omega t + \pi/3$ and get:

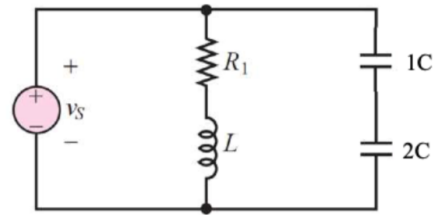
$$\begin{aligned} x_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T 4 \cos^2(\alpha) + 24 \cos(\alpha) + 36 dt} \\ &= \sqrt{\frac{1}{T} \left(4 \int_0^T \cos^2(\alpha) dt + 24 \int_0^T \cos(\alpha) dt + 36 \int_0^T dt \right)} \\ &= \sqrt{\frac{1}{T} \left(4 \frac{T}{2} + 24 \cdot 0 + 36T \right)} \\ &= \sqrt{\frac{1}{T} (2T + 36T)} \\ &= \sqrt{38}. \end{aligned}$$

$$\begin{aligned} \langle x(t) \rangle &= 6 \\ x_{\text{rms}} &= \sqrt{38}. \end{aligned}$$

RESULT

Problem 2B

Find the equivalent impedance of the circuit shown on the Figure on the right.



Derivation

The impedance of a resistor Z_R is simply its resistance:

$$Z_R = R.$$

For the inductor the impedance Z_L is

$$Z_L = \omega L \angle \frac{\pi}{2} = j\omega L.$$

Hence, the total impedance of the middle branch is:

$$Z_m = R + j\omega L.$$

For the right branch we have two capacitors in series, which we will replace with an equivalent capacitor:

$$C_{eq} = \frac{C \cdot 2C}{C + 2C} = \frac{2}{3}C.$$

The impedance of a capacitor Z_C is simply

$$Z_C = \frac{1}{\omega C} \angle -\frac{\pi}{2} = \frac{1}{j\omega C}.$$

Which means the equivalent impedance of the right branch is:

$$Z_r = Z_{C_{eq}} = \frac{1}{j\omega \frac{2}{3}C} = \frac{3}{2j\omega C}.$$

Now we combine the impedances of the middle branch Z_m and the right branch Z_r using the formula for equivalent parallel impedance:

$$\begin{aligned} Z_{eq} &= \frac{1}{\frac{1}{Z_m} + \frac{1}{Z_r}} \\ &= \frac{Z_m Z_r}{Z_m + Z_r} \\ &= \frac{(R + j\omega L) \left(\frac{3}{2}j\omega C\right)}{R + j\omega L + \frac{3}{2}j\omega C} \\ &= \frac{3(R_1 + j\omega L)}{2j\omega C(R_1 + j\omega L) + 3}. \end{aligned}$$

$$Z_{eq} = \frac{3(R_1 + j\omega L)}{2j\omega C(R_1 + j\omega L) + 3}$$

RESULT